



SECP2523: DATABASE

SEM I 2025/ 2026

**DATABASE DESIGN DESCRIPTIONS: CONCEPTUAL DATABASE
DESIGN
FOR HASTA VEHICLE RENTAL MANAGEMENT SYSTEM**

Group Name: DaTeam

Members:

- 1. MUHAMMAD ADAM ASHRAFF BIN ZAMRI (GROUP LEADER)**
- 2. MIKAEL HAQIMI BIN NAHAR JUNAIDI**
- 3. AHMAD WILDAN BIN MAZANI**
- 4. JOSHUA LING SHANG YANG**

Stakeholder: HASTA TRAVEL & TOURS SDN. BHD

Representative Name	Date Interviewed
1. ALIF FIRDAUS BIN JALALUDIN	15 th DECEMBER 2025

Printing Date: 4 February 2026

TABLE OF CONTENTS

CHAPTER	TOPIC	PAGE
1	Introduction	
	1.1 Overview about the company	Page 2
	1.2 Problems	Page 3
	1.3 Proposal	Page 4
	1.4 Definitions, Acronyms and Abbreviations	Page 6
2	DFD (to-be)	
	2.1 DFD Level 0	Page 7
	2.2 DFD Level 1	Page 8
3	Data & Transaction Requirement	
	3.1 Proposed Business Rule	Page 10
	3.2 Proposed Data & Transactional	Page 13
4	Database Conceptual Design	
	4.1 Conceptual ERD	Page 22
	4.2 Enhanced ERD(EERD)	Page 23
5	Data Dictionary	Page 24
6	Summary	Page 34
7	Appendices	Page 35

1. Introduction

1.1. Overview about the company

Hasta Travel & Tours Sdn. Bhd. is a licensed service provider for travel and transportation services based in Skudai, Johor. The organization founded by Nabihah Kamarudin alumni of UTM and began as a student movement in August 2018. However, the organization has grown and become a significant contributor to the Universiti Teknologi Malaysia community. Hasta Travel is currently located at the Universiti Teknologi Malaysia Student Mall and provides services in the following areas: car rentals, travel agency services, and logistics.

Hasta Travel is designed to accommodate students, lecturers, and the general public. They also offer car rental services with their array of vehicles, ranging from economical hatchback cars such as Perodua Axia/Myvi to sedans, SUVs, and MPVs.

The mission of Hasta Travel is to facilitate convenient transportation services for UTM students and travellers in Johor by offering cheap and safe transportation. And the vision is to be one of the leading student-centric travel service brands in Malaysia by filling the gap between academic and professional travel services. Their promise is to deliver safety, excellence in vehicle maintenance, and support for local university programs.

1.2. Problems

1.2.1 Heavy Reliance on WhatsApp

Hasta Travel still depends a lot on WhatsApp for booking communication. Even though customers fill in a form that sends their details to WhatsApp automatically, staff must handle the rest of the process manually through chat. This causes slow confirmations, inconsistent communication, and delays when staff are unavailable.

1.2.2 Limited Real-Time Management in DriveO

DriveO is an online system that provides minimal control measures for the staff. Bookings cannot be processed in real time since the staff is still expected to confirm the vehicle availability. Usually, this causes more confusion and even booking of a single vehicle by multiple customers simultaneously.

1.2.3 Lack of Important Features

The company has customer profiles, but there is basic customer information, which does not include any loyalty or reward programs. Only notifications are sent via WhatsApp, which may be missed by the customer. Data security is low, and the reports provided by the system are not complex enough to analyze the performance of the system.

1.2.4 The workflow is split between two platforms

Staff must switch between WhatsApp (for customer communication) and DriveO (for record keeping). This split workflow slows down operations, makes coordination harder and increases the chance of human error.

1.2.5 Poor Accuracy and Customer Experience

The current system makes it hard to maintain accurate information, respond quickly to customers, and track vehicles efficiently. As a result, customers do not receive a smooth and professional experience during the booking process.

1.2.6 The process does not scale well for a large target market.

With an estimated 30,000 potential users (mainly university students), the current WhatsApp-first workflow will become overwhelming, making it hard to respond quickly and maintain high service quality.

1.3. Proposal

The proposed solution for Hasta Travel is a centralized and device-agnostic vehicle rental management system that aims to automate existing manual processes and provide a seamless digital experience for users. By integrating various business operations into one system, the proposed system aims to address existing business operations and provide better services to the UTM community.

1.3.1 Digital Lifecycle Management

The system will manage the entire lifecycle of the rental process, moving away from fragmented manual entries towards a smooth and synchronized digital lifecycle. This entire change can be divided into three main phases:

- **Pre-Rental & Verification**

The system will introduce a secure and digital onboarding process, where customers will have the ability to upload their identity documents (Student ID and Driving License). The system will then provide automatic updates when the staff verifies the customer's profile.

- **Booking & Financial Automation**

Customers will have the ability to search and filter the vehicle according to the availability. The system will provide cost estimates instantly and will also manage the deposit pre-authorizations. The system will also provide a digital agreement and will allow the customer to pay through QR code transfer or online transfer.

- **Post-Rental & Loyalty**

Once the customer returns the vehicle, the system will calculate the final charges and will automatically deduct the late return fees, if applicable. To ensure customer retention, the system will provide a loyalty program that will allow customers to earn points on every successful rental.

1.3.2 Enhanced Operational Control

For the staff and the administrators, the solution offers control at the highest level through the provision of a dedicated management dashboard.

- **Fleet & Maintenance Tracking**

The solution tracks the miles covered by the vehicles, eliminating the need for physical logs, with the ability to send reminders for maintenance.

- **Role-Based Management**

The solution offers access control, thereby enhancing data security. The staff are able to concentrate on the day-to-day activities, such as verification of documents and inspection of vehicles, while the Admin can access the financial reports, demand trends, and risk management tools, including the ability to blacklist clients.

- **Quality Assurance**

The provision of the feedback and rating tool enables the company to track the quality of service provided, thereby enhancing data-driven improvements based on actual customer feedback.

1.4. Definitions, Acronyms and Abbreviations

Definitions:

- Customer - The targeted users of the system (Students & Staff in Universiti Teknologi Malaysia)
- Staff - The workers of the Hasta Travel & Tours Sdn. Bhd. They could access all the features of the management side except Staff Management.
- Administrator - Also used in short term “Admin”, who are the main people that own the company of Hasta Travel & Tours Sdn. Bhd. The administrator has full access to the system without any restrictions.
- Hasta - The short term used for representing “Hasta Travel & Tours Sdn. Bhd.”
- Loyalty points - One of the extra features for the system. Customers could get these points from the rental bookings and use the points to redeem rewards and vouchers.
- Voucher - A redeemable-code that could be used during the booking phases to obtain a discount for the booking with lower price.

Acronyms:

- ERD - Entity-Relationship Diagram
- DFD - Data Flow Diagram
- DB - Database
- ID - Identity Document
- UTM - Universiti Teknologi Malaysia
- FAQ - Frequently Asked Questions
- PK - Primary Key
- FK - Foreign Key
- NRIC - National Registration Identity Card
- IC - Identity Card

2. DFD (to-be)

2.1 DFD Level 0

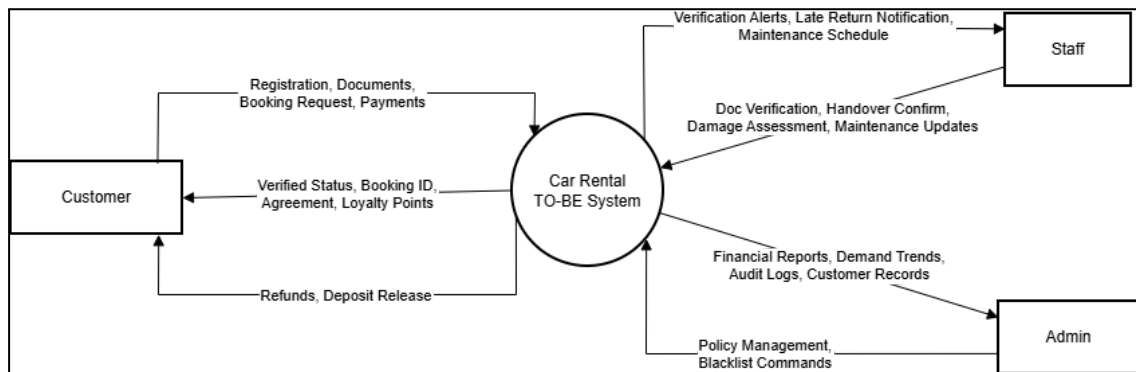


Figure 1: DFD Level 0

Figure 1 shows the Context Diagram (Level 0) provides a high-level overview of the car rental system and identifies the main data flows among the system and entities. The Customer provides registration documents and booking requests, receiving verified status updates, digital agreements and loyalty rewards in return. The Staff responsible for document verification and vehicle maintenance updates. Meanwhile the system provides late returns auto alert and tasks verification. Also, the Admin is able to access system policies and blacklists, gaining access to financial reports, audit logs and customer records to ensure smooth operations.

2.2 DFD Level 1

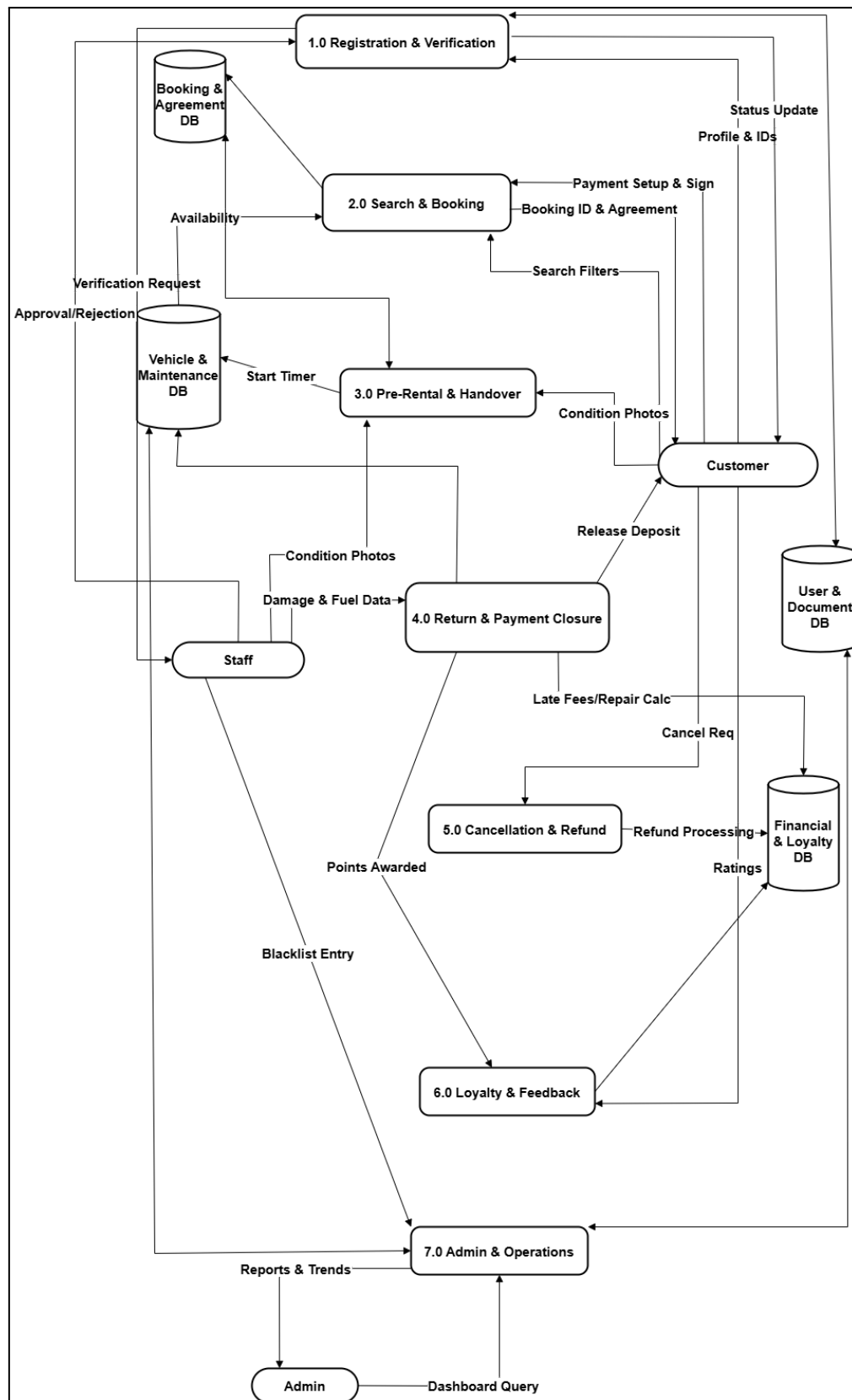


Figure 2: DFD Level 1

Figure 2 visualizes the Level 1 Data Flow Diagram (DFD) where data interacts between customers, staff and administrators within the to-be system. The system workflow starts with user registration and identity verification where validated profiles are stored into the User & Document DB. Customers then search for vehicles and finalize bookings, creating records in the Booking Agreement DB while referencing real-time data from the Vehicle & Maintenance DB. During the rental period, the system tracks vehicle handovers, returns, and damage assessments, calculating fees and loyalty points that are managed within the Financial & Loyalty DB. Finally, the diagram shows how administrative oversight is maintained through a centralized dashboard create business trend reports, financial records and blacklists user.

3. Data & Transaction Requirement

This section will describe the way in which the Hasta Vehicle Rental Management System processes data and the specific rules that the database must follow in order to ensure that the information is accurate, secure, and useful to the business.

3.1. Proposed Business Rule

To turn this current manual method of operation into a professional digital system, we have developed a number of ground rules which the software automatically enforces. This acts as a digital supervisor, ensuring that all individuals renting a vehicle from Hasta Travel are qualified to do so and that vehicles are not misplaced. In this way, many human mistakes can be avoided, such as double-booking or failing to collect a deposit, which currently causes confusion and inefficiency for Hasta Travel. The following are the policies which ensures a fair, safe, and efficient system for all parties concerned:

3.1.1. User Account & Eligibility

3.1.1.1. Verified Access Only: Only after a customer's Student ID Card & Driving License had been verified by a staff member would the customer be eligible to book a vehicle.

3.1.1.2. Unique Account Integrity: Only one account per customer using a unique email & phone number to organize records & avoid duplication.

3.1.1.3. Document Validity Rule: Customers would not be eligible to register or book vehicles if their driving license had expired.

3.1.1.4. Secure Document Storage: To ensure customer data privacy & data security requirements, customer identity documents would be stored in an encrypted form.

3.1.1.5. Blacklisting: If customers had abused the system or damaged vehicles on multiple occasions, they would be blacklisted by the staff.

3.1.2. Booking & Reservation Logic

3.1.2.1. Real-Time Availability: The system will only display vehicles that are available in real time for the dates entered to ensure that no two individuals book a vehicle at the same time.

3.1.2.2. Vehicle Status Lock: When a vehicle is marked as “Rented”, it will automatically be removed from the search list until the return process is complete.

3.1.2.3. Debt-Free Booking: If a customer has an outstanding fine or penalty for a previous rental, they will not be allowed to make bookings until this debt is cleared.

3.1.2.4. Agreement Requirement: A digital agreement must be signed by the customer before completing a booking to ensure legal recourse in case of an issue.

3.1.2.5. Booking Cancellation Window: A customer can only make changes to a booking within a certain timeframe to receive a refund.

3.1.3. Financial & Payment Procedures

3.1.3.1. Financial Integrity Rule: A handover can only be confirmed if a successful deposit payment proof has been recorded.

3.1.3.2. Payment Receipt Verification: Manual payment methods such as QR and online transfers need a staff member to verify the receipt uploaded before the booking can be fully confirmed.

3.1.3.3. Deposit Refund Policy: Deposits are refunded when all rental charges, penalties, and damage charges are calculated.

3.1.3.4. Voucher Redemption Limit: Voucher redemption can only be applied to some bookings as per management rules.

3.1.4. Vehicle Handover & Inspection

3.1.4.1. Pre-Rental Inspection Proof: The customer/staff (optional) needs to inspect the condition of the car and send a photo before handing over the keys.

3.1.4.2. Vehicle Handover Confirmation: The rental timer will begin counting down only after a staff confirms the physical pickup.

3.1.4.3. Post-Rental Damage Assessment: The staff needs to inspect before handing over the deposit to the customer and see if they got any new damages.

3.1.5. Fleet & Operational Management

3.1.5.1. Maintenance Warnings: The system will be able to monitor the number of miles the car has traveled and alert the team about a safety check or oil change.

3.1.5.2. Fleet Balance Monitoring: The system will have to offer information about the pickup and return schedule to prevent flooding the staff.

3.1.5.3. Automated Notification: The system will have to offer information to the customer about their booking status, verification results, and payment confirmations sent via email.

3.1.6. Reporting & Quality Assurance

3.1.6.1. Staff Only Data Control: Only staffs and admin can access the financial data and customer's personal information.

3.1.6.2. Role-Based Reporting: Only Admin can access the financial KPI report and revenue report.

3.1.6.3. Loyalty Milestones: The system may record the number of times the customer makes a booking and reward the customer with special discounts.

3.1.6.4. Feedback Loop: The customer may give feedback regarding the service after each time he rents.

3.2. Proposed Data & Transactional

This specifies exactly what information the system needs to collect and how it processes that information to run the business efficiently.

3.2.1 Data Entry

The system collects the following detailed information to facilitate full accountability in all modules:

3.2.1.1 User & Profile Details

- **Basic Info:** Full Name, Email Address, Secure Password, Mobile Phone Number.
- **Address Info:** Full Residential Address/Hostel Address.
- **Identification Info:** Upload Student/Staff ID Card image & Driving License image.
- **Emergency Contacts:** Emergency Contact Name & Contact Phone Number.
- **Profile Visuals:** Personal Profile Picture for verification purposes.

3.2.1.2 Vehicle & Fleet Records

- **Identification:** Vehicle Plate Number, Brand, and specific Model name.
- **Specifications:** Manufacturing Year, Exterior Color, and Fuel Type.
- **Technical Info:** Transmission Type (Auto/Manual), Seating Capacity, and current Odometer/Mileage reading.
- **Pricing:** Configurable Hourly Rental Rate and Daily Rental Rate for 1, 3, 5, 7, 9, 12 and 24 hours.
- **Documents:** Digital uploads of Vehicle Grant, Insurance policies, and Road Tax documents.
- **Visuals:** High-quality images of the vehicle from multiple angles.

3.2.1.3 Booking & Rental Process

- **Core Data:** Booking ID generated by system, Customer ID, Vehicle ID.
- **Timeline:** Pickup Date and Time, Return Date and Time.
- **Financials:** Calculated Total Rental Price and Required Deposit Amount.
- **Legal:** Signed Rental Agreement file.
- **Notes:** Special comments or customer requests about the booking.

3.2.1.3 Financial & Payment Evidence

- **Payments:** Unique Transaction ID and Payment Method (Online Transfer/QR).
- **Verification:** Digital upload of the bank transfer or QR payment receipt.
- **Deposits:** Records of Deposit holds and subsequent Deposit Refund status.
- **Fines:** Specific records for late return penalties or damage fees.

3.2.1.4 Inspection, Maintenance & Feedback

- **Condition Logs:** Pre-rental and Post-rental inspection photos to track car state.
- **Levels:** Recorded Fuel levels and Odometer readings at pickup and return.
- **Damages:** Detailed descriptions of any new scratches, dents, or mechanical issues.
- **Maintenance:** Service Date, Service Type, and maintenance cost.
- **Reviews:** Customer Star Ratings and written Feedback/Comments after rental.

3.2.1.5 Loyalty & Reward Data

- **Reward Rules:** Definition of reward names, required milestones, and discount types.
- **Points Entry:** Number of points awarded per successful rental transaction.
- **Voucher Generation:** Unique Voucher Codes, expiry dates, and their assigned values.
- **Redemption Records:** Tracking which Voucher ID was used for which Booking ID.

3.2.2 Data Update

The system is designed to allow staff and administrators to maintain a high level of data accuracy. Based on the system's operational logic, here are all the possible data updates that can be performed:

3.2.2.1 Vehicle & Fleet Maintenance

- **Availability Status:** Switching a vehicle's status from "Available", "Booked", "Rented", and "In Maintenance" by manual or automatic means.
- **Odometer Tracking:** After each rental trip, current mileage of a vehicle is updated to keep the records accurate.
- **Technical Specifications:** Changing the vehicle information like the rental charge (hourly or daily), color, or transmission type in case of an upgrade in the fleet.
- **Document Renewal:** Updating the images of vehicle insurance, road tax, and grant renewal on a yearly basis.

3.2.2.2 User Account & Safety Management

- **Verification Workflow:** Updating the status of a new user from 'Pending' to 'Verified' or 'Rejected' upon reviewing their documents.
- **Account Restrictions:** Updating the status of a customer to 'Blacklisted' or 'Restricted' in cases of fraud, damage, or non-payment of fees.
- **Reason Documentation:** Updating particular reasons for rejecting documents or blacklisting accounts to communicate with the customer.
- **Profile Maintenance:** Allowing users to update their personal contact details, home addresses, or security passwords.
- **Role Adjustments:** Allowing administrators to update staff access levels or employee details through the management module.

3.2.2.3 Booking Lifecycle Management

- **Schedule Adjustments:** Modifying the pickup or return dates and times if a customer requests a rental extension.
- **Operational Stages:** Manually moving a booking through its lifecycle: “Pending Approval”, “Confirmed”, “Picked Up” and “Completed”.
- **Cancellation Processing:** Marking a booking as “Cancelled” and updating the system to trigger refund or vehicle release logic.
- **Location Updates:** Changing the designated pickup or return location for specific booking records.
- **Remarks & Notes:** Adding internal staff remarks or special customer requests to an existing booking file.

3.2.2.4 Financial & Penalty Finalization

- **Payment Verification:** Updating the status of a rental payment from “Pending” to “Successful” once the receipt is verified.
- **Penalty Status:** Changing penalty or fine records from “Unpaid” to “Paid” after confirming the customer has settled the debt.
- **Fee Adjustments:** Adding or modifying extra charges such as fuel shortages, cleaning fees, or minor repair costs during the return inspection.
- **Deposit Release:** Updating the status of a security deposit to “Released” or “Refunded” after a successful car return.

3.2.2.5 Engagement & Loyalty Updates

- **Reward Point Balance:** Customer’s total loyalty point is automatically updated after a successful rental completion.
- **Voucher Status:** If a customer redeems a voucher, the status changes to “Used” or “Expired”.
- **Service Improvements:** Quality standards are updated based on customer ratings and feedback comments.

3.2.3 Data Queries/Outputs

The system provides a wide range of automated outputs, views, and downloadable reports designed to give each user role exactly the information they need without manual searching.

3.2.3.1 Customer Queries (Self-Service Information)

- **Real-Time Vehicle Search:** Finding cars that are currently available based on specific pickup dates, return dates, and preferred car models.
- **Active Booking Status:** Checking the real-time progress of a current rental, such as whether it is “Confirmed”, “Picked Up” or “Overdue”.
- **Personal Rental History:** Retrieving a full list of all past completed, cancelled, or rejected bookings.
- **Digital Documents:** Accessing and downloading the signed digital rental agreement and the final booking invoice.
- **Loyalty & Rewards Summary:** Check current balance of loyalty points, history of earnings, and list of active vouchers.
- **Financial Statements:** View breakdown of total costs paid, held security deposits, and paid penalty fees or any outstanding.
- **Profile Status:** Check if identity documents have been verified or if the account needs updating.

3.2.3.2 Staff Queries (Operational Coordination)

- **Identity Verification Queue:** Listing all new customer registrations that are waiting for staff to review and approve uploaded IDs and licenses.
- **Booking Approval Dashboard:** Keeping an eye on the new booking requests that either need manual confirmation or verification of payment receipts.
- **Daily Fleet Schedule:** Getting a filtered list of the vehicle pick-ups and returns scheduled for the same day to ready the cars.
- **Vehicle Status Overview:** Checking which cars in the fleet are currently “Available”, “Rented”, “Reserved”, or “In Maintenance”.

- **Customer Risk Profiles:** The system tracks customer history of late returns, damaged items, and unpaid fines to determine their eligibility for blacklisting.
- **Maintenance Alerts:** The system identifies vehicles which have reached their service mileage thresholds. It also identifies vehicles which have reached their designated maintenance dates.
- **Inventory Details:** Accessing full technical records, insurance expiry dates, and road tax status for every car in the fleet.

3.2.3.3 Admin Queries (Management & Strategy)

- **Total Revenue Reports:** Preparing financial statements showing total income from rental, penalty, and other fees over the course of different weeks or months.
- **Deposit Tracking:** Keeping an eye on the amount of security deposits the system holds and which are planned to be released.
- **Booking Trend Analysis:** Mapping out the demand patterns to determine the most popular days or car models among UTM students.
- **Fleet Performance:** Checking the vehicle usage and maintenance cost records to decide.
- **Staff Performance & Audit:** Checking the staff members' actions such as verification approvals and booking confirmations.
- **System Usage Statistics:** Keeping track of the number of active users, new registrants, and overall system growth.
- **Customer Feedback Summary:** Loving ratings and comments posted by users to help service areas identify where the service needs to be improved.

Table 1: Proposed Data and Transactional

Category	Data Entry (Inputs)	Data Updates	Data Queries (Outputs)
User & Profile	Name, Email, UTM ID/License images, emergency contacts, profile photos.	Verification status (Verified/Rejected), blacklisting for policy violations, profile info updates.	Profile verification status, customer risk profiles for staff, identity verification queues.
Vehicle & Fleet	Plate number, model, technical specs, rental rates, and legal documents (grant/tax).	Availability status (Available/Rented/Maintenance), odometer tracking, and annual document renewal.	Real-time vehicle search, fleet availability overview, and maintenance/service alerts.
Booking & Rental	Booking ID, timestamps, total price, deposit amount, and signed digital agreements.	Rental lifecycle stages (Confirmed/Picked Up/Completed), extensions, and cancellation processing.	Active booking status (e.g., Overdue), rental history, and digital rental agreements for download.
Finance & Penalties	Transaction IDs, payment receipts (QR/Transfer), penalty records, and deposit hold logs.	Payment verification status, penalty finalization (Paid/Unpaid), and deposit release logic.	Financial statements, revenue reports (Total/Weekly/Monthly), and deposit tracking.

Category	Data Entry (Inputs)	Data Updates	Data Queries (Outputs)
Loyalty & Rewards	Reward rules, points awarded per rental, voucher codes, and redemption logs.	Automated reward point balances and voucher status changes (Used/Expired).	Loyalty point balances, history of earnings, and active/claimable vouchers.
Inspection & Feedback	Pre/Post-rental condition photos, fuel/mileage levels, and customer ratings/reviews.	Quality standard adjustments and detailed logs of damages or mechanical issues.	Customer feedback summaries, service quality ratings, and vehicle usage/maintenance costs.

Table 1 above represents a brief summary of the data architecture of the system, based on the categorization of the data into three main areas of functionality, namely what is inputted into the system (Data Entry), what is done to maintain the data (Data Updates), and what is outputted from the system (Data Queries). It briefly describes how the system replaces the current manual system of WhatsApp coordination by creating an organized digital system, where all the data, whether used to identify the student who carries it or the car's maintenance history, is used to improve the speed of operation, financial accuracy, and user experience within the UTM community.

4. Database Conceptual Design

4.1. Conceptual ERD

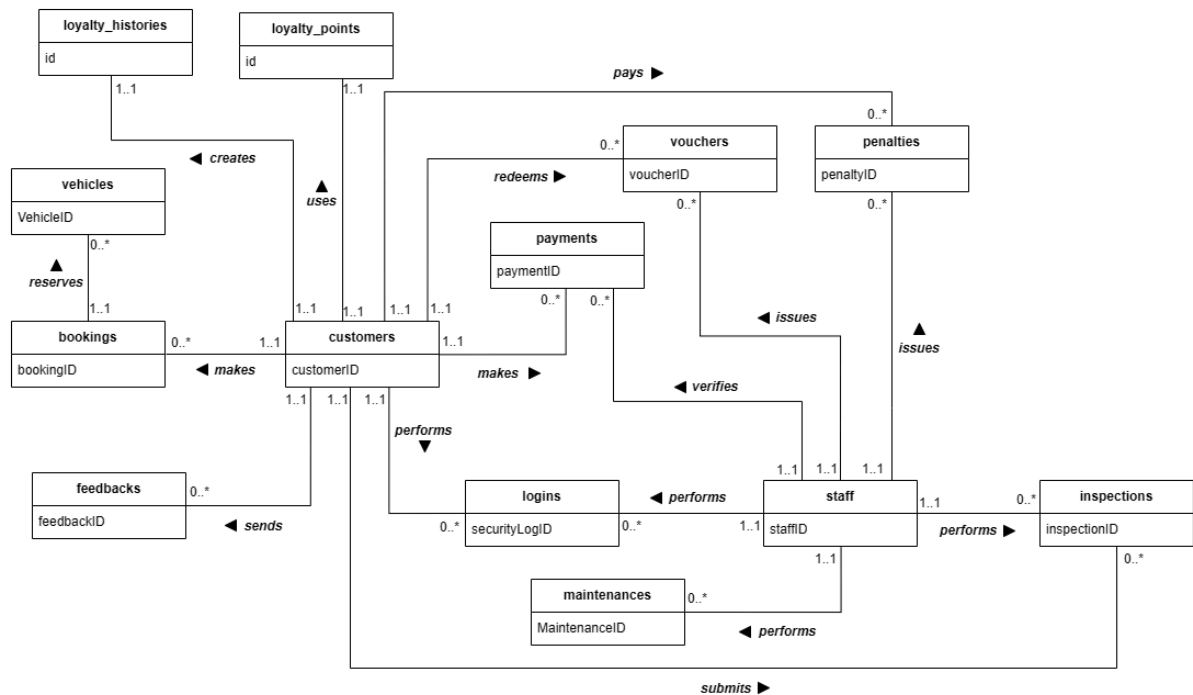


Figure 3: Conceptual ERD

This Conceptual ERD is showing every entity included in our system. And for each entity, there is a unique key, which is the primary key as shown as in **Figure 3**. There is also the relationship between the entities to show how the entities interact for development of the entire system.

4.2. Enhanced ERD(EERD)

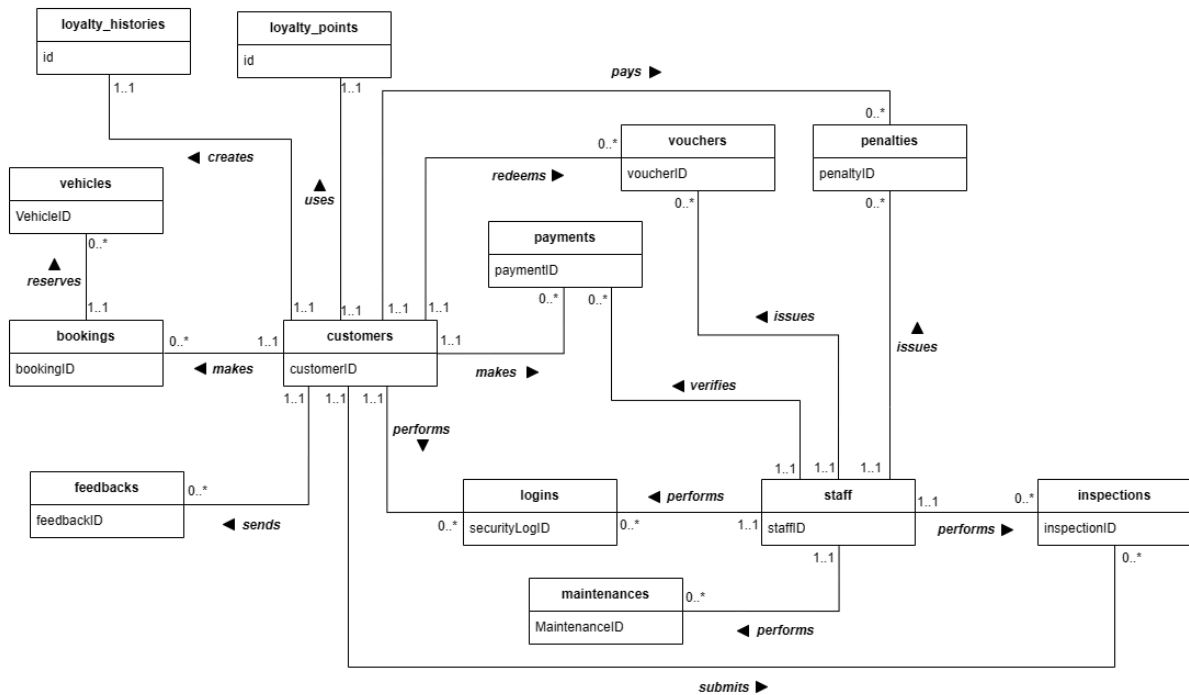


Figure 4: EERD

The EERD will be the modification of the conceptual ERD if there are any changes in the diagram. Since the conceptual ERD in Figure 3 is already finished and completed, the EERD will be the same as the conceptual ERD as shown in **Figure 4**.

5. Data Dictionary

Table 2 shows detailed information about the data used for the conceptual ERD. It shows entity name, attributes, data type & length, nulls, description, and example of the data. This table is able to improve understanding of the data used in the system.

Table 2: Data Dictionary for the conceptual ERD

Entity Name	Attributes	Description	Example
staff	staffID (PK)	Unique identifier for staff (Primary Key).	1
	name	Full name of the staff member.	Admin User
	role	Access role (e.g., Admin, Staff).	Admin
	email	Unique login email.	admin@dateam.com
	phoneNo	Contact number for operations.	012-3456789
	password	Hashed security password.	(hashed_string)
	active	Account status flag.	TRUE
customers	customerID (PK)	Unique identifier for customer (PK).	1001
	fullName	Full legal name.	Sarah Ali
	email	Unique login email.	sarah@utm.my
	phoneNo	Contact number.	019-8765432
	stustaffID	Student or Staff Matric ID.	A19EC0001
	driving_license_expiry	Driving license expiration date.	2028-05-20
	ic_passport	NRIC or Passport number.	990101-01-5555

Entity Name	Attributes	Description	Example
	dob	Date of Birth.	1999-01-01
	nationality	Nationality.	Malaysia
	faculty	Faculty or Department name.	Computing
	homeAddress	Permanent home address.	123 Jalan Skudai
	collegeAddress	Residential college address.	Kolej Tun Dr Ismail
	bankName	Bank name for refunds.	Maybank
	bankAccountNo	Bank account number.	164012345678
	emergency_contact_name	Name of emergency contact person.	Ali Bin Abu
	emergency_contact_no	Phone number of emergency contact.	012-9998888
	accountStat	Account status (Active, Pending).	Active
	previous_account_stat	Status held before blacklisting.	Active
	blacklisted	Flag indicating if user is banned.	FALSE
	blacklist_reason	Reason for blacklisting.	Damaged vehicle
	rejection_reason	Reason for application rejection.	Invalid License
	avatar	Path to profile picture file.	profile/1001.jpg
	student_card_image	Path to student card image.	docs/card.jpg

Entity Name	Attributes	Description	Example
	ic_passport_image	Path to IC/Passport image.	docs/ic.jpg
	driving_license_image	Path to driving license image.	docs/license.jpg
vehicles	VehicleID (PK)	Unique identifier for vehicle (PK).	501
	plateNo	Registration plate number (Unique).	JKV 8822
	model	Vehicle model name.	Myvi 1.5 AV
	type	Body type (e.g., Compact, SUV).	Compact
	vehicle_category	Broad category ('car' or 'bike').	car
	brand	Manufacturer brand.	Perodua
	year	Manufacturing year.	2023
	color	Vehicle color.	Red
	priceHour	Base rental rate per hour.	8.00
	hourly_rates	Tiered pricing structure.	{"1": 8, "24": 100}
	availability	Current availability status.	TRUE
	mileage	Current mileage counter.	45000
	owner_name	Name of vehicle owner.	Hasta Travel
	owner_phone	Contact number of owner.	011-2223333
	owner_nric	NRIC/ID of owner.	800101-01-1111

Entity Name	Attributes	Description	Example
	road_tax_image	Path to road tax document.	vehicles/tax.jpg
	grant_image	Path to grant document.	vehicles/grant.jpg
	insurance_image	Path to insurance document.	vehicles/ins.jpg
	blocked_dates	Dates vehicle is unavailable.	["2025-12-25"]
bookings	bookingID (PK)	Unique identifier for booking (PK).	2005
	customerID (FK)	Foreign Key to Customer.	1001
	vehicleID (FK)	Foreign Key to Vehicle.	501
	staffID (FK)	Foreign Key to Staff (Approver).	1
	voucherID (FK)	Foreign Key to Voucher.	801
	bookingDate	Date the booking was created.	2025-12-01
	originalDate	Scheduled start date.	2025-12-05
	bookingTime	Scheduled start time.	10:00:00
	returnDate	Scheduled return date.	2025-12-07
	returnTime	Scheduled return time.	10:00:00
	pickupLocation	Location for vehicle pickup.	KTDI Parking

Entity Name	Attributes	Description	Example
	returnLocation	Location for vehicle return.	KTDI Parking
	totalCost	Total calculated rental cost.	150.00
	bookingStatus	Current status (e.g., Completed).	Completed
	bookingType	Type (Standard, Extended).	Standard
	invoiceLink	Path/Link to PDF Invoice.	inv/2005.pdf
	aggreementLink	Path/Link to Agreement PDF.	agrmt/2005.pdf
	remarks	Optional customer requests.	Need baby seat
payments	paymentID (PK)	Unique identifier for payment (PK).	3001
	bookingID (FK)	FK to Booking (Nullable).	2005
	amount	Amount paid in transaction.	150.00
	depoAmount	Security deposit amount held.	50.00
	transactionDate	Date and time of payment.	2025-12-01 14:30
	paymentMethod	Method (e.g., QR, Cash).	DuitNow
	paymentStatus	Status (e.g., Paid, Pending).	Paid
	depoStatus	Deposit status (Held/Refunded).	Refunded

Entity Name	Attributes	Description	Example
penalties	penaltyID (PK)	Unique identifier for penalty (PK).	4001
	bookingID (FK)	FK to Booking (Nullable).	2005
	customerID (FK)	FK to Customer.	1001
	amount	Total penalty amount.	20.00
	reason	Reason for the penalty.	Late Return
	date_imposed	Date/Time penalty was issued.	2025-12-07 12:00
	lateReturnHour	Hours returned late (if any).	2
	penaltyFees	Specific fee for lateness.	20.00
	penaltyStatus	Payment status (Unpaid/Paid).	Unpaid
	payment_proof	Path to receipt image.	penalties/rec.jpg
	paid_at	Timestamp when paid.	2026-01-08 10:00
maintenance	MaintenanceID (PK)	Unique identifier (PK).	901
	VehicleID (FK)	FK to Vehicle being serviced.	501
	StaffID (FK)	FK to Staff overseeing work.	1
	description	Details of maintenance work.	Oil Change
	cost	Cost incurred.	120.00

Entity Name	Attributes	Description	Example
	date	General date of service.	2025-12-20
	start_time	Precise start time.	2025-12-20 09:00
	end_time	Precise completion time.	2025-12-20 11:00
	type	Type (maintenance, holiday, etc).	maintenance
	reference_id	External reference ID.	REF-12345
vouchers	voucherID (PK)	Unique identifier (PK).	801
	customerID (FK)	Foreign Key to Customer.	1001
	user_id (FK)	Alias for customerID (FK).	1001
	voucherCode	Unique code for redemption.	SDKTN-8821
	code	Alias for voucherCode.	SDKTN-8821
	voucherAmount	Fixed discount amount.	5.00
	discount_percent	Percentage discount value.	20
	voucherType	Type (Rental Discount, Merchant).	Merchant Reward
	validFrom	Start of validity period.	2026-01-01
	validUntil	End of validity period.	2026-03-01
	expires_at	Alias for validUntil.	2026-03-01

Entity Name	Attributes	Description	Example
	conditions	Terms of use.	Min spend RM50
	terms_conditions	Alias for conditions.	Min spend RM50
	isUsed	Usage flag.	FALSE
	status	Status string (e.g. 'unused').	unused
	redeem_place	Location to redeem.	Sedap Kitchen
loyalty_points	id (PK)	Unique identifier (PK).	601
	user_id (FK)	Foreign Key to Customer.	1001
	points	Current points balance.	500
	tier	Tier (Bronze, Silver, Gold).	Gold
	rental_bookings_count	Qualified bookings count.	5
loyalty_histories	id (PK)	Unique identifier (PK).	950
	user_id (FK)	Foreign Key to Customer.	1001
	points_change	Points added (+) or deducted (-).	+100
	reason	Transaction description.	Rental Reward
rewards	id (PK)	Unique identifier (PK).	701
	name	Name of the reward.	SEDAP KITCHEN
	offer_description	Details of the offer.	RM 5 OFF
	points_required	Points cost to redeem.	200

Entity Name	Attributes	Description	Example
	category	Category (Food, Milestone).	Food
	code_prefix	Prefix for voucher codes.	SDKTN
	milestone_step	Booking count trigger step.	12
	discount_percent	Discount percentage value.	50
inspections	inspectionID (PK)	Unique identifier (PK).	1101
	bookingID (FK)	FK to Booking.	2005
	staffID (FK)	FK to Staff.	1
	inspectionType	Pre-rental or Post-rental.	Post-rental
	damageCosts	Estimated cost of damages.	0.00
	photosAfter	Photos after rental (JSON/Path).	["img1.jpg"]
	photosBefore	Photos before rental (JSON/Path).	["img0.jpg"]
feedback	feedbackID (PK)	Unique identifier (PK).	1201
	customerID (FK)	Foreign Key to Customer.	1001
	bookingID (FK)	Foreign Key to Booking.	2005
	rating	Rating score (e.g., 1-5).	5
	comment	Textual feedback.	Great service!

Entity Name	Attributes	Description	Example
	type	Feedback type (Complaint/Review).	Review
	status	Status (Pending/Reviewed).	Reviewed

6. Summary

This Conceptual Design Report phase addresses key inefficiencies experienced in the company's current business process. One of the issues faced is strong reliance on a working process between WhatsApp for communication and the DriveO system for record-keeping. This split workflow has led to slow booking confirmation, a lack of real-time vehicle availability and unable to serve their target market, which is a large market base for the UTM community.

The proposed solution will be a central vehicle rental management system that will automate the entire customer experience. Firstly, the management system is digitalised where the system will automate the process through a centralized management system which will be composed of a Pre-Rental & Verification process, a Booking and Financial Automation process and a Post-Rental and Loyalty process.

The next solution is enhancing operational control which the system will be controlled through a set of proposed business rules that will be used to ensure data integrity and user eligibility. The user that is eligible for completing the booking process is for the verified access only where customers must be verified through documents provided to the company's staff to be eligible to book a vehicle.

Also, the user can experience the real-time availability feature. This feature prevents any booking of a vehicle for a specific time slot when it has already been allocated to another customer. The other interesting feature is debt-free booking. This ensures that a customer who has any pending penalties will not be allowed to make a new booking and claim any rewards. The financial integrity rule will authorize payment where the proof of deposit payment is required for a successful Handover.

Finally, for the overview of this phase the Data Architecture outlines specific requirements for Data Entry, Data Update and Data Query for key areas such as User & Profile, Vehicle & Fleet, Booking and Rental, and Financial and Penalties. This digital structure, aided by a detailed Data Dictionary, replaces the manual system with an organized, efficient and scalable platform.

7. Appendices

7.1. Use Case Diagram for the system

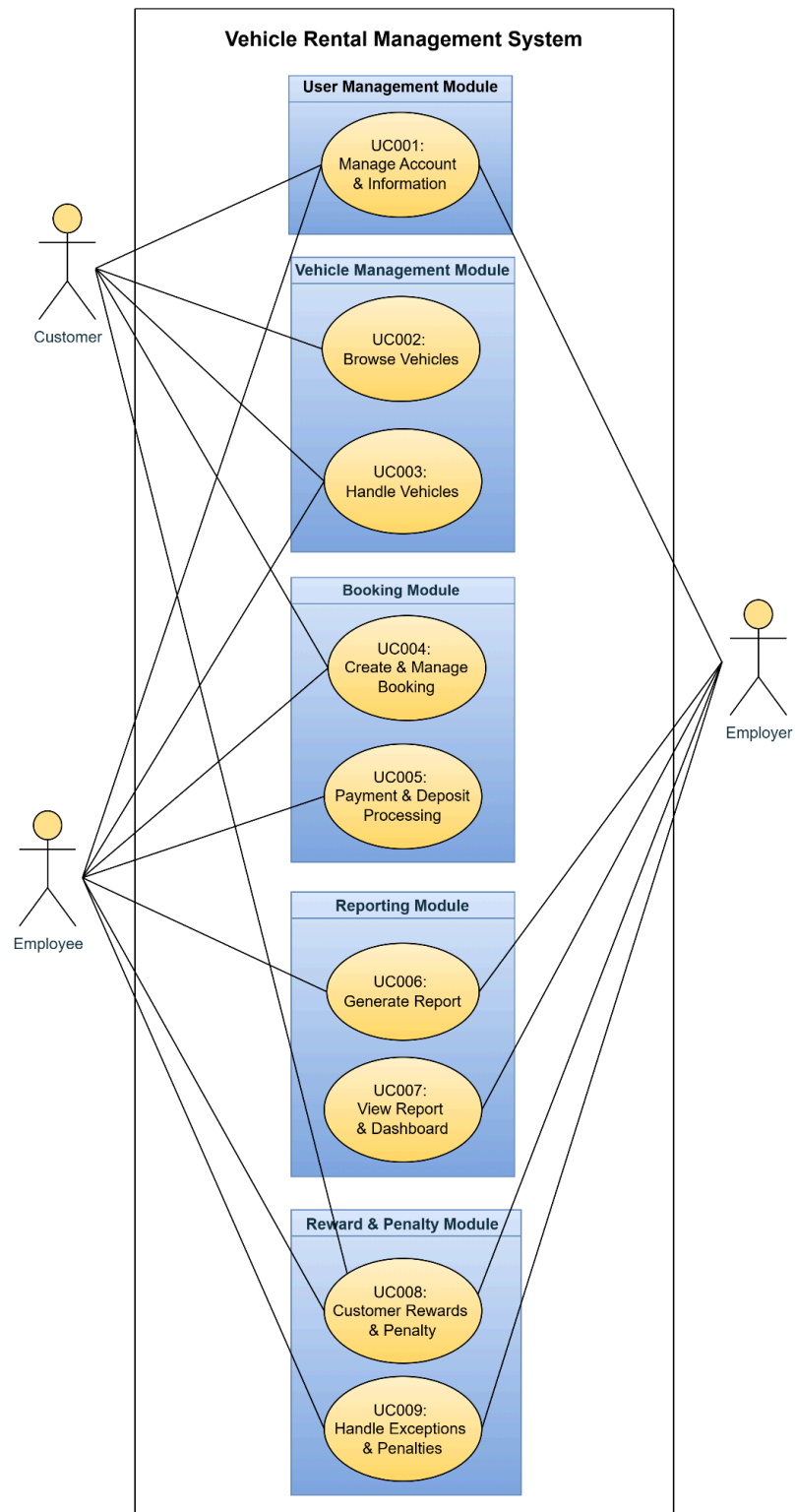


Figure 5: Use Case Diagram for Vehicle Rental Management System

7.2. Activity Diagram for every use case identified

7.2.1. UC001

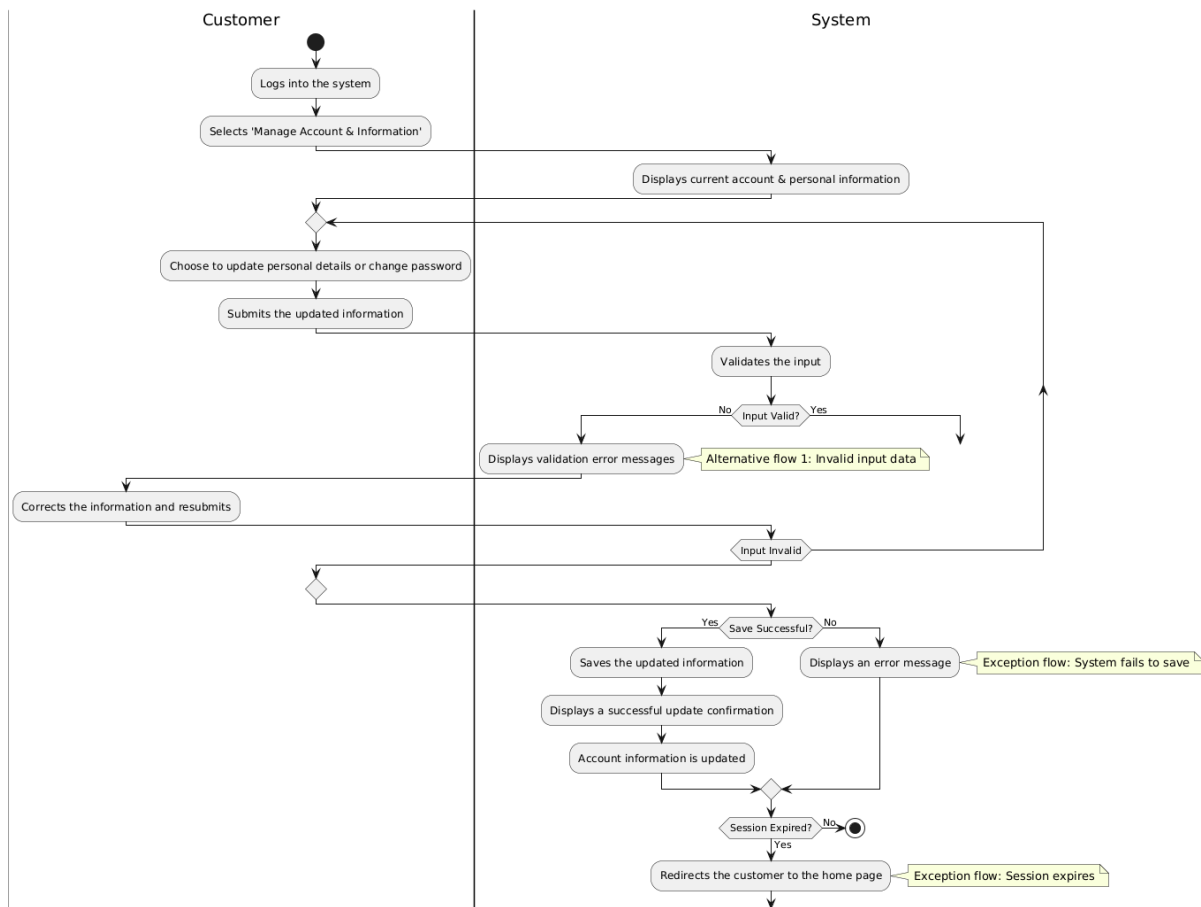


Figure 6: Activity Diagram for Manage Account & Information

Figure 6 details the chronological interactions between the Customer, System, and Database required to update account information. It highlights the sequential steps of retrieving current data, validating user inputs, and confirming the update in the database while handling potential exceptions.

7.2.2. UC002

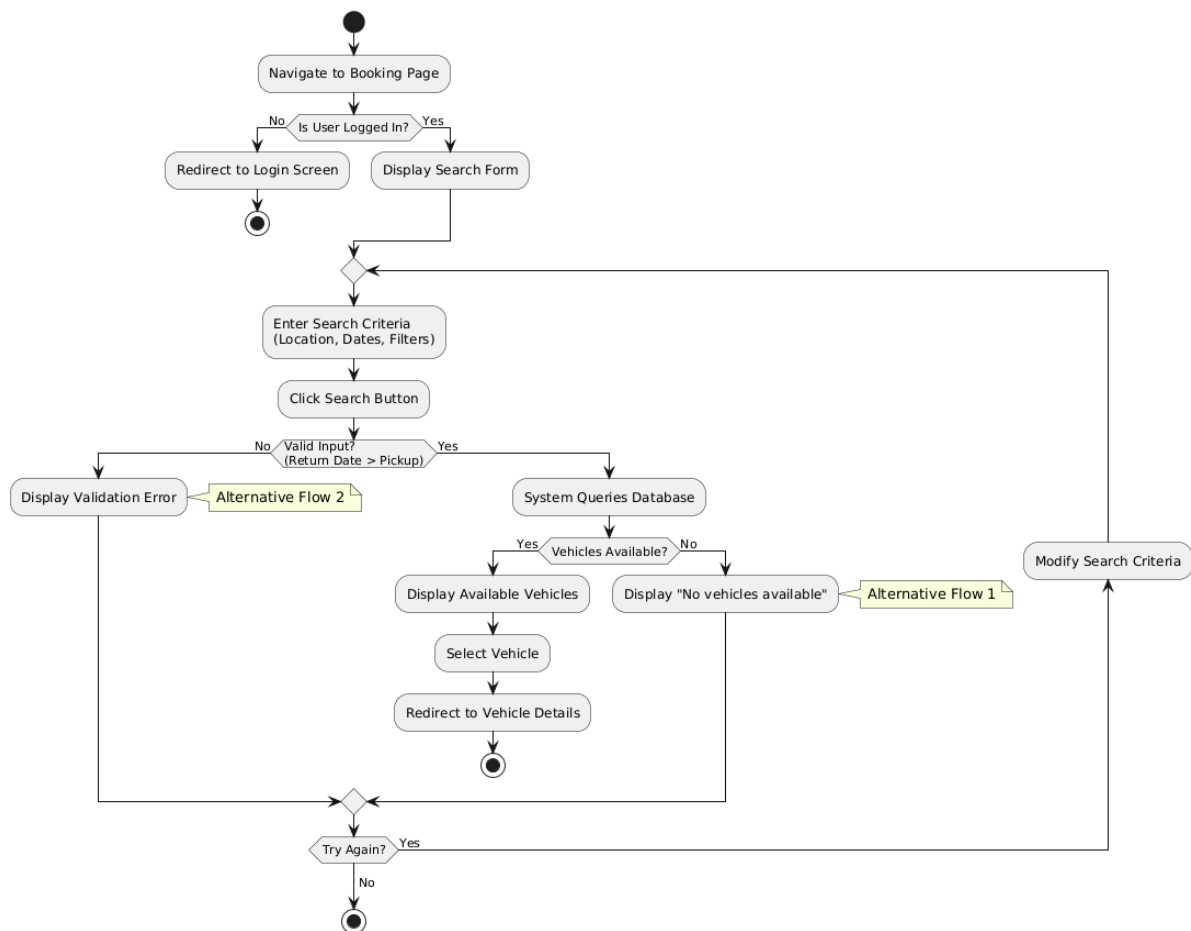


Figure 7: Activity Diagram for Browse Vehicles

Figure 7 describes the activity workflow for browsing cars, which starts with a log in prior to gaining access to the searching interface. Successful completion of this activity requires that a user chooses an available car from among the list created and is then taken to another webpage for details.

7.2.3. UC003

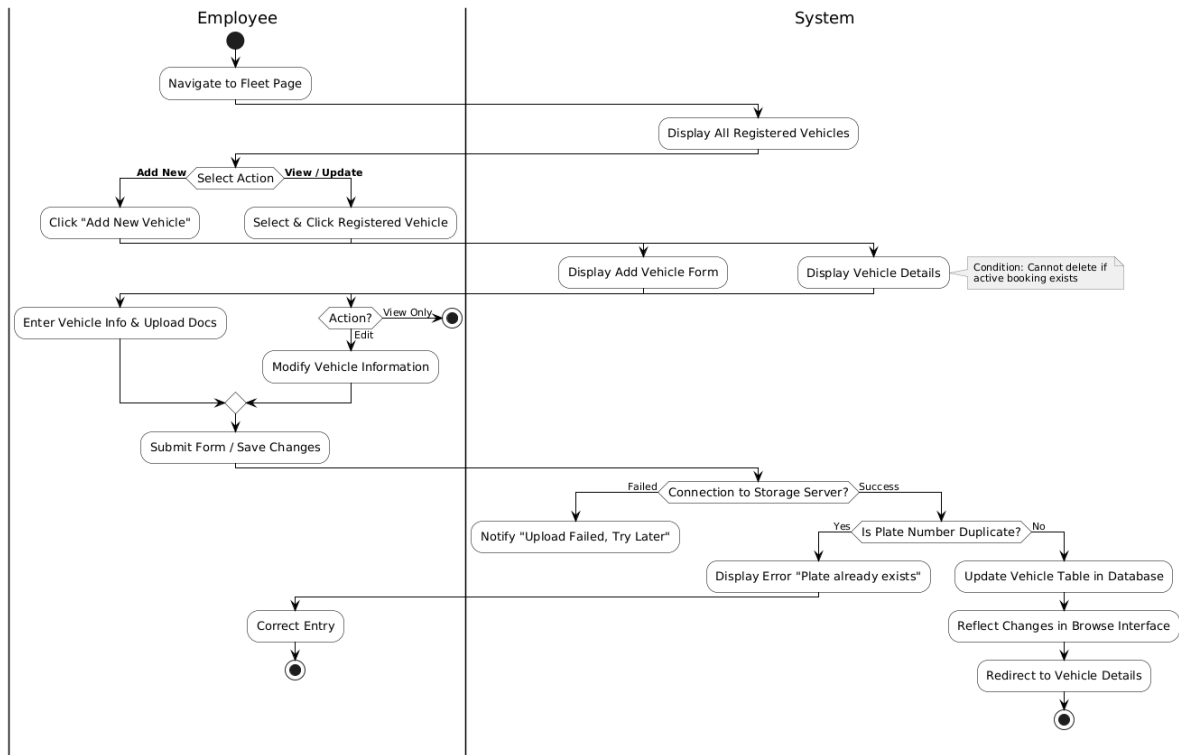


Figure 8: Activity Diagram for Handle Vehicles

Based on **Figure 8**, it shows system interaction that clearly details the message sequence flow from the Staff, Fleet UI, Vehicle Controller, to the Database. The system has broken the process into two different operation flows, Main Flow, which deals with the display and update of current data in vehicles, while Alternative Flow 1 deals with adding new inventory to the system. Error handling is also shown in the system interaction through alternative flows.

7.2.4. UC004

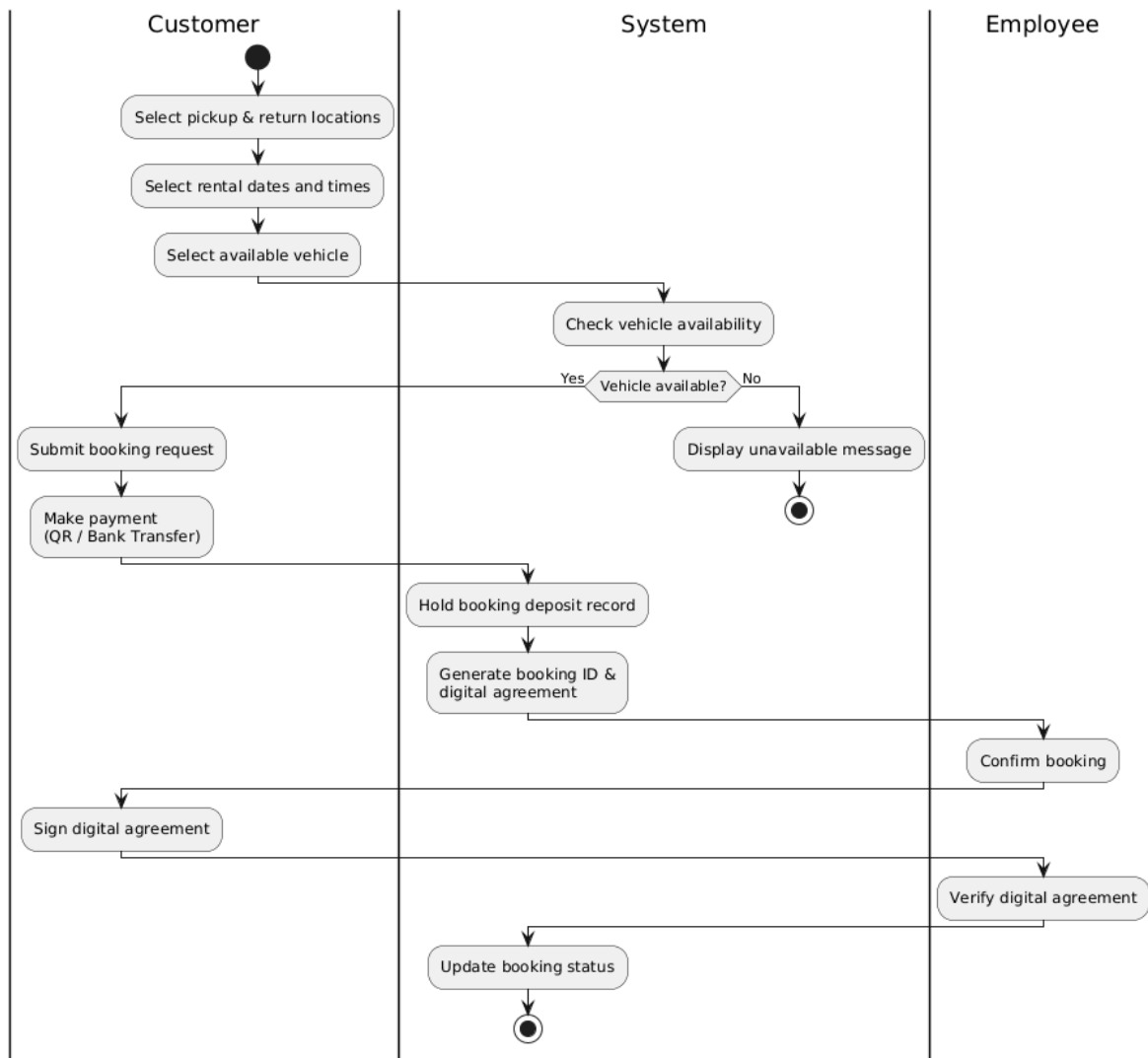


Figure 9: Activity Diagram for Manage Account & Information

Figure 9 illustrates a sequence diagram of the customer's booking process. The customer interacts with BookingPage and BookingSystem while The Employee interacts with PaymentSystem. The system will display unavailable message if the vehicle is not available.

7.2.5. UC005

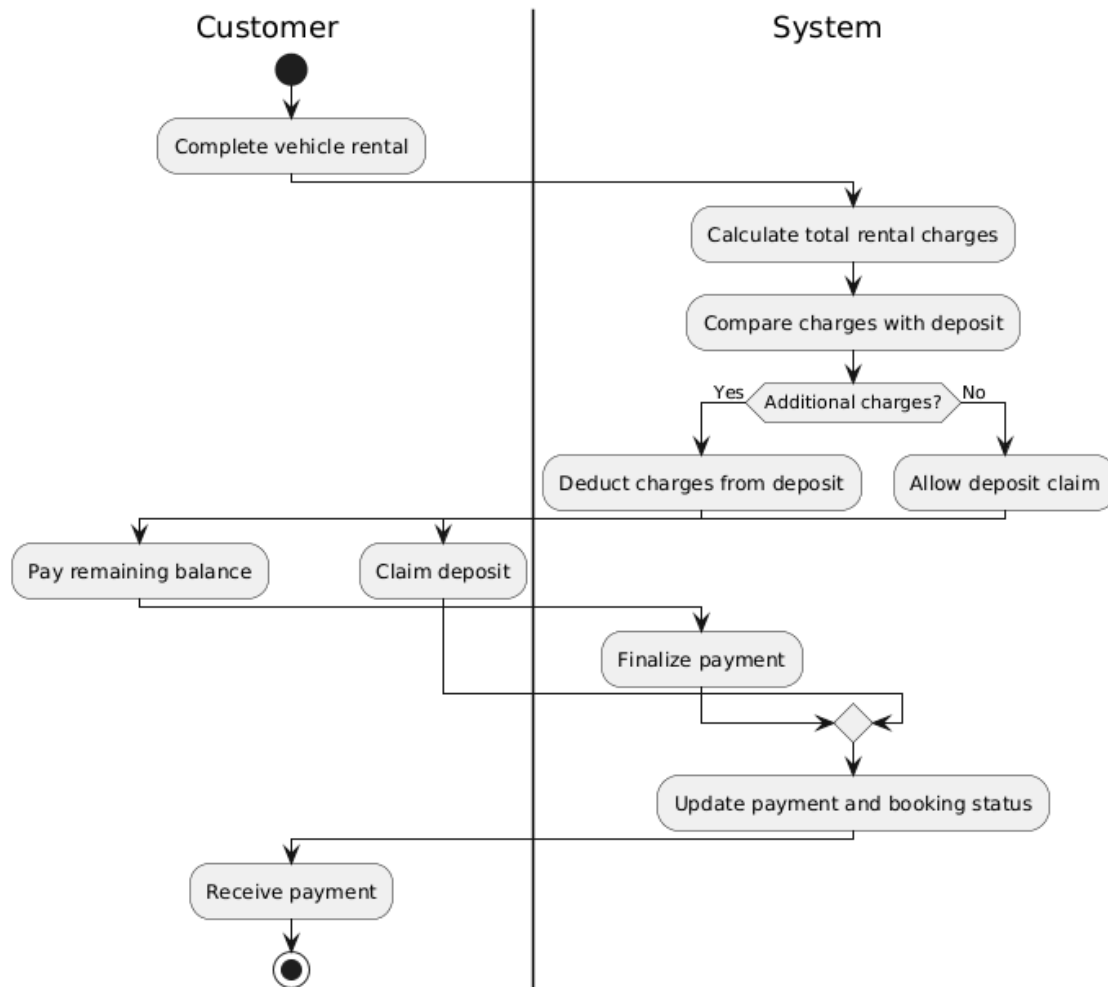


Figure 10: Activity Diagram for Payment & Deposit Processing

Figure 10 illustrates the sequence diagram for the payment processing process happened between Payment System and Booking System. The process begin when complete vehicle rental and the booking system calculate total rental charges with deposit held if there is problem with the booking the system will deduct charges from deposit and request for remaining balance, if there is no problem with the booking the system will allow deposit claims.

7.2.6. UC006

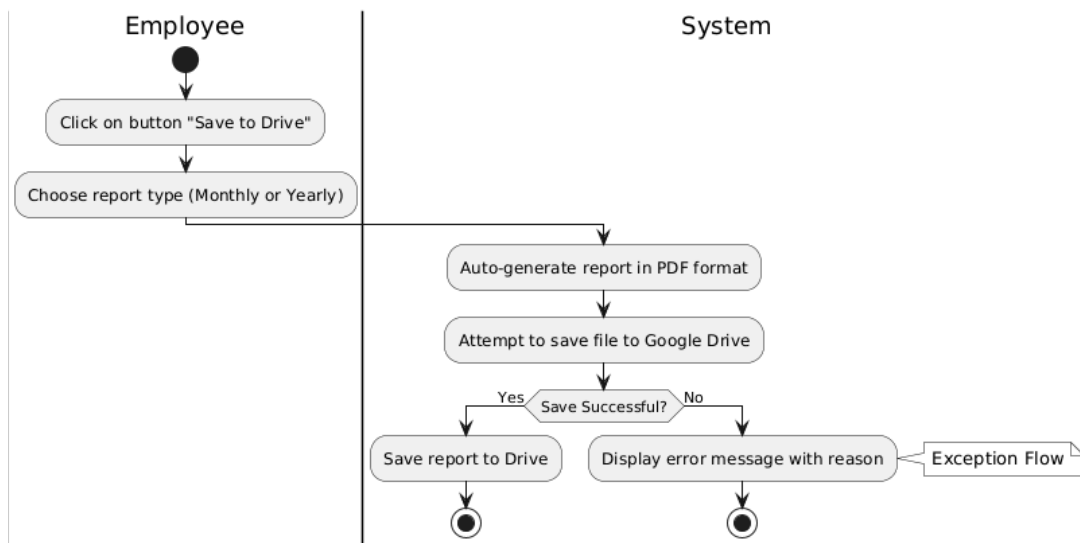


Figure 11: Activity Diagram for Generate Report

Figure 11 illustrates the employee's workflow for creating reports. It shows the process of selecting a report type (monthly or yearly) and the system's automated actions to generate the PDF and save it directly to Google Drive.

7.2.7. UC007

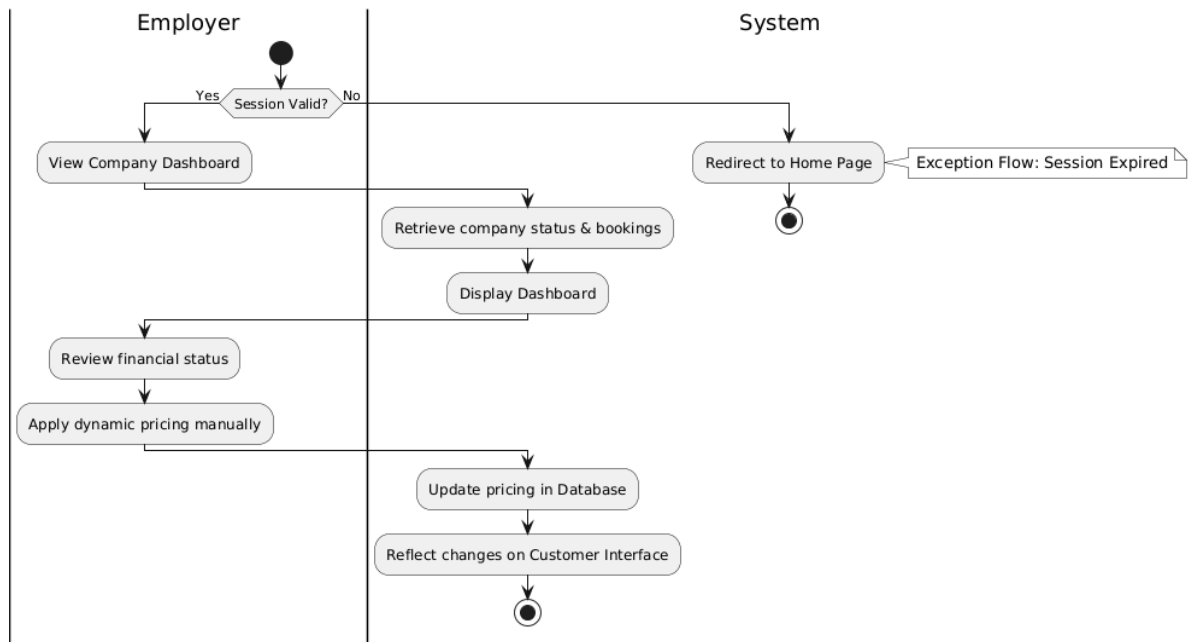


Figure 12: Activity Diagram for View Review & Dashboard

Figure 12 maps the employer's interaction with the company dashboard. It visualizes the flow of viewing real-time financial status and bookings, as well as the logic for manually applying dynamic pricing updates to the system.

7.2.8. UC008

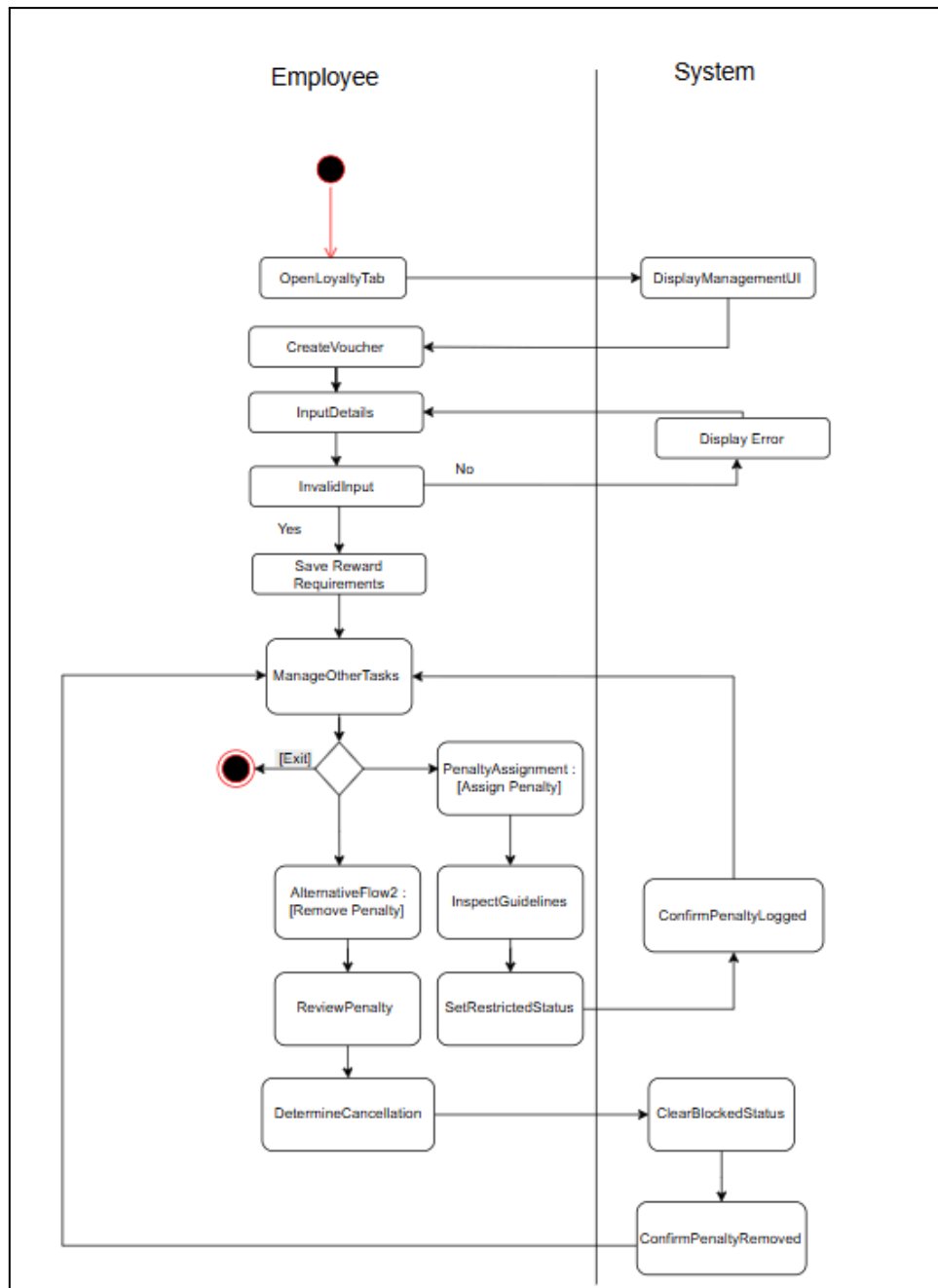


Figure 13: Activity Diagram for Customer Rewards & Penalty

Figure 13 illustrates the flow of exchange of data required to manage loyalty and penalty features within the dashboard. It depicts the Employee initiating tasks like voucher creation while the System validates inputs and updates the backend status. The workflow specifically highlights the transition of saving reward requirements where the System subsequently logs or clears penalties and updates user restrictions based on the Employee's real-time decisions.

7.2.9. UC009

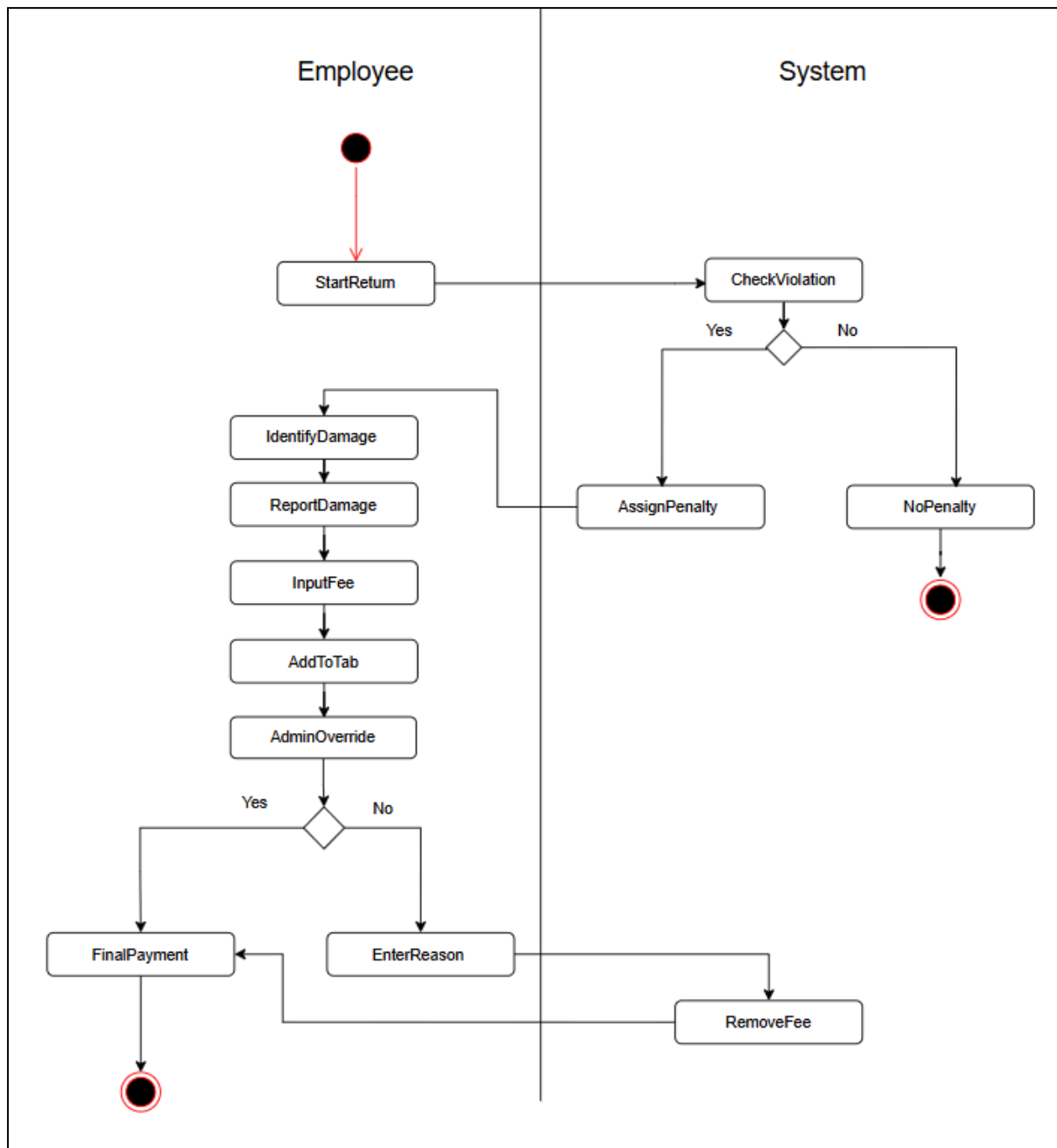
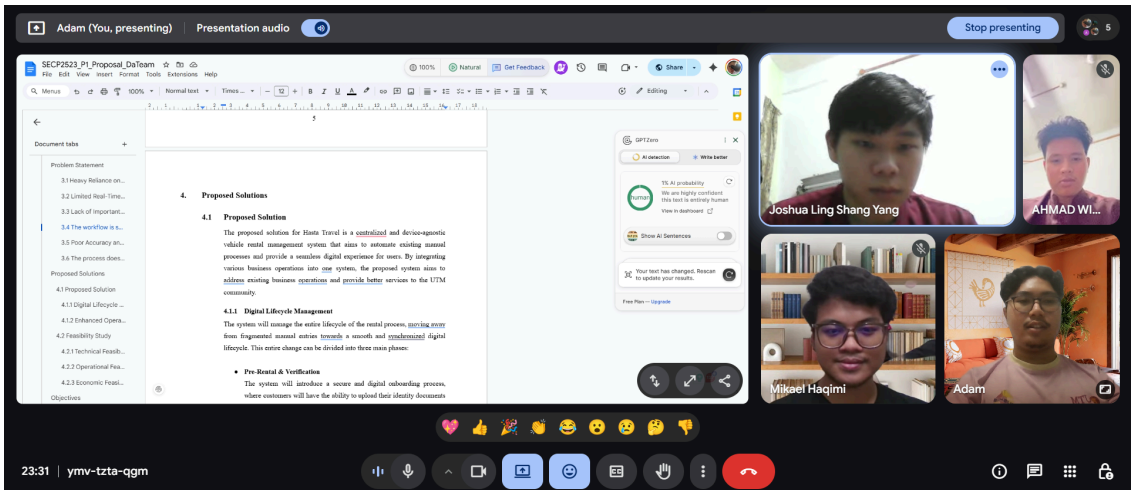


Figure 14: Activity Diagram for Handle Exceptions & Penalties

Figure 14 illustrates the chronological exchange of data required to handle vehicle returns and manage potential damage charges. It depicts the **System** executing an automated violation check, which subsequently prompts the **Employee** to assess the vehicle for damages and input applicable fees.

7.3. Meeting Log

Meeting Minute #1

DATE/TIME	24 th November 2025		
LOCATION	Google Meet (Online Meeting)		
AGENDA	Task Distribution and Finalize Report		
MEETING MC	MUHAMMAD ADAM ASHRAFF BIN ZAMRI		
ATTENDANCE			
NAME	TIME	REASON FOR ABSENCE	
MUHAMMAD ADAM ASHRAFF BIN ZAMRI	11.30 AM	-	
MIKAEL HAQIMI BIN NAHAR JUNAIDI	11.30 AM	-	
AHMAD WILDAN BIN MAZANI	11.30 AM	-	
JOSHUA LING SHANG YANG	11.30 AM	-	
GROUP PHOTO			
			
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTION AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Task Distribution	• Adam distributes all	• Adam

		tasks evenly.	
2	Task Discussion	• All members read and identify the requirements for this task.	• All members
3	Finalize Reports	• All members recheck all information and discuss possible improvements.	• All members
4	Meeting ended	• Adam ended the meeting at 2.30 PM	• Adam